Workplace Fire Safety

Module 1: Preventing and responding to fires

1 Introduction

Most workplace fires are preventable. This module explains **how a fire starts**, which **extinguisher** fits which fire, and what every employee does when the alarm sounds. It takes about 20 minutes and ends with a short set of learning checks.

1.1 Learning objectives

After this module you can:

- Name the three elements a fire needs and explain how removing one stops the fire.
- Match the classes of fire to their typical fuels.
- Choose a suitable extinguisher and use it with the PASS technique.
- Follow the evacuation procedure of your building.

2 How a fire starts

A fire needs **heat**, **fuel** and **oxygen** at the same time. Remove any one of the three and the fire goes out: water removes heat, a fire blanket removes oxygen, and closing a gas valve removes fuel.

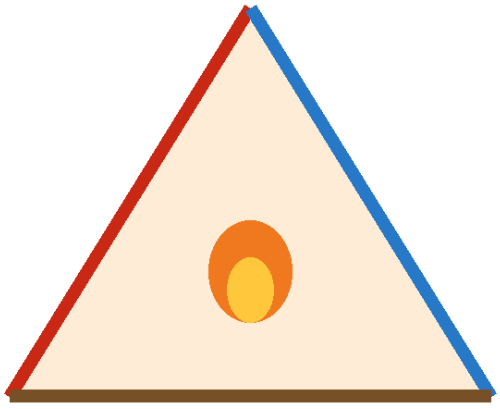

Figure 1: The fire triangle. The sides stand for heat (red), oxygen (blue) and fuel (brown).

2.1 Classes of fire

Fires are grouped by the fuel that burns. The class tells you which extinguisher is safe to use.

Class	Fuel	Typical examples in an office or workshop
A	Solid materials	Paper, cardboard, wood, textiles, plastics
B	Flammable liquids	Solvents, paint thinners, fuel, cleaning alcohol
C	Flammable gases	Propane cylinders, natural gas lines, aerosol cans
D	Combustible metals	Magnesium and aluminium swarf from machining
F	Cooking oils and fats	Deep fryers and pans in staff kitchens

Class	Fuel	Typical examples in an office or workshop
Electrical	Live equipment (not a class of its own)	Overheated chargers, extension leads, server racks, distribution boards
Mixed	More than one fuel	Waste bins with paper and solvent rags, storage rooms

Note: Never use water on a fire that involves live electrical equipment or cooking oil. Switch off the power first if it is safe to do so.

3 Responding to a fire

When you discover a fire, act in this order:

1. Raise the alarm by pressing the nearest manual call point.
2. Call the emergency services and give the address of the building.
3. Decide whether the fire is small enough to fight safely.
4. If it is not, close the doors behind you and leave the building.

3.1 Using an extinguisher

Use an extinguisher only on a small fire and only if your escape route stays behind you. Remember the **PASS** technique: **Pull** the pin, **Aim** at the base of the fire, **Squeeze** the handle and **Sweep** from side to side.

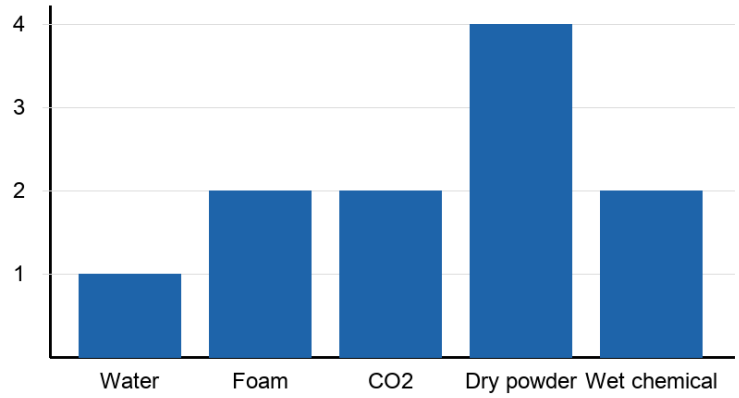

Figure 2: Number of fire classes each extinguisher type is suitable for.

Extinguisher	Label colour	Use on	Do not use on
Water	Red	Class A	Electrical, B, F
Foam	Cream	Classes A and B	Electrical, F
CO2	Black	Class B, electrical	Class F, enclosed rooms
Dry powder	Blue	Classes A, B, C, electrical	Indoors where possible
Wet chemical	Yellow	Classes A and F	Electrical, B

Remember: Your safety comes first. Only fight a fire if you have been trained and the fire is no bigger than a waste bin.

4 Evacuation

Every building has an evacuation plan with **escape routes** and an **assembly point**. Read the plan near the entrance of your floor.

- Leave by the nearest safe exit and do not use the lifts.
- Help visitors who do not know the building.
- Report to the fire warden at the assembly point.
- Do not go back into the building until the fire service says it is safe.